Victor Nikhil Antony

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Education

Doctor of Philosophy in Computer Science, *Johns Hopkins University* **August 2021 — Present**

- Advised by Dr. **Chien-Ming Huang**
- ACM/IEEE HRI Pioneer 2025

Master of Science in Engineering (Computer Science), *Johns Hopkins University* **May 2024**

Bachelor of Science in Computer Science, University of Rochester **May 2021**

- Advised by Dr. **Ehsan Hoque** and Dr. **Zhen Bai**

Publications

Under Review/Under Preparation

Victor Nikhil Antony*, Shiye Cao*, Shuning Wang and Chien-Ming Huang – ELLA: Generative AI-powered Social Robots for Early Language Development **[Under review at IDC'26]** *equal contribution **2026**

Victor Nikhil Antony, Zhili Gong, Yoonjae Kim and Chien-Ming Huang – Introducing M: A Modular, Modifiable, Low-Cost Socially Assistive Robotics Research Platform **[Under review at IROS'26]** **2026**

Victor Nikhil Antony, Kai Liang, Travolta Chiposi and Chien-Ming Huang – Robots As Mindfulness Anchors: Learnings from Participatory Design and Home Deployments **[Manuscript under preparation]** **2026**

Shuning Wang, Shiye Cao, **Victor Nikhil Antony,** and Chien-Ming Huang – On Detecting and Mitigating Child-Robot Interaction Disengagement in Home Environments **[Manuscript under preparation]** **2026**

Published

Victor Nikhil Antony, Adithya R N, Sarah Derrick, Zhili Gong, Peter Donley and Chien-Ming Huang – Plant-inspired Robot Design Metaphors for Ambient HRI. *IEEE/ACM International Conference on Human-Robot Interaction (HRI'26)* **2026**

Victor Nikhil Antony*, Zhili Gong*, Guanchen Li, Clara Jeon and Chien-Ming Huang – *Lantern: A Minimalist Robotic Object Platform. IEEE/ACM International Conference on Human-Robot Interaction (HRI'26)* **2026**

Victor Nikhil Antony, Clara Jeon, Jiasheng Li, Ge Gao, Huiashu Peng, Anastasia Ostrowski and Chien-Ming Huang - *The Design of On-Body Robots for Older Adults. IEEE/ACM International Conference on Human-Robot Interaction (HRI'25)* **2025**

Victor Nikhil Antony, Maia Stiber and Chien-Ming Huang - *Xpress: Generating Dynamic, Context-Aware Robot Facial Expressions Using Language Models. IEEE/ACM International Conference on Human-Robot Interaction (HRI'25)* **2025**

Amama Mahmood, Shiye Cao, Maia Stiber, **Victor Nikhil Antony**, and Chien-Ming Huang - *Voice Assistants for Health Self-Management: Designing for and with Older Adults. ACM CHI conference on Human Factors in Computing Systems (CHI'25)* **2025**

Victor Nikhil Antony, Mengchi Li, Shu-Han Lin, Junxin Li and Chien-Ming Huang - *Social Robots for Sleep Health: A Scoping Review.* International Journal of Social Robotics **2025**

Sally Cao**,** Jiwon Moon, Amama Mahmood, **Victor Nikhil Antony**, Ziang Xiao, Anqi Liu, and Chien-Ming Huang **–** *Interruption Handling for Conversational Robots. Robotics: Systems and Science (RSS'25)***.** **2025**

Victor Nikhil Antony and Chien-Ming Huang - *ID. 8: Co-Creating visual stories with Generative AI*. ACM Transactions on Interactive Intelligent Systems (TIIS) **2024**

Ulas Karli, Juo-Tung Chen, **Victor Nikhil Antony** and Chien-Ming Huang - *Alchemist: LLM-Aided End-User Development of Robot Application*. *ACM/IEEE International Conference on Human-Robot Interaction(HRI'24)* **2024**

Victor Nikhil Antony*, Sue Min Cho* and Chien-Ming Huang - *Co-designing with older adults, for older adults: robots to promote physical activity*. ACM/IEEE International Conference on Human-Robot Interaction*(HRI'23)* **2023**

Kurtis Haut, Caleb Wohn, **Victor Nikhil Antony,** Aidan Goldfarb, Melissa Welsh, Dillanie Sumanthiran, M Rafayet Ali, Ehsan Hoque - *Demographic feature isolation for bias research using deepfakes*. ACM International Conference on Multimedia **2022**

Wasifur Rahman, Sangwu Lee, Md Saiful Islam, **Victor Nikhil Antony**, Harshil Ratnu, Mohammad Rafayet Ali, Abdullah Al Mamun, Ellen Wagner, Stella Jensen-Roberts, Emma Waddell, Taylor Myers, Meghan Pawlik, Julia Soto, Madeleine Coffey, Aayush Sarkar, Ruth Schneider, Christopher Tarolli, Karlo Lizarraga, Jamie Adams, Max A Little, E Ray Dorsey, Ehsan Hoque - *Detecting Parkinson's disease using a web-based speech task: Observational study*. Journal of medical Internet research **2021**

Workshop Publications

Victor Nikhil Antony, Shiye Cao and Chien-Ming Huang - *Practical Trade-offs in Generative Robot Expression Design*. HRI 2026 Vizij Workshop+Tutorial on Expressive Robot Faces **2026**

Kai Liang, **Victor Nikhil Antony,** Junxin Li, Hao Su and Chien-Ming Huang - Examining Barriers to Equitable Exosuit Use by Older Adults. HRI 2026 Equitable Robotics for Wellbeing Workshop **2026**

Victor Nikhil Antony and Chien-Ming Huang - *Designing Social Robots that Engage Older Adults in Exercise: A Case Study*. HRI 2024 Workshop on HRI for Aging in Place **2024**

Victor Nikhil Antony, Adira Blumenthal, Ziyue Qiu, Ashely Tenesaca, Wanyin Hu, Zhen Bai - *Projection-Based AR for Hearing Parent-Deaf Child Communication*. Frameless **2020**

Research/Work Experience

Research Assistant, *Johns Hopkins University, Intuitive Computing Lab* **August 2021 — Present**
- Building socially assistive robots to deliver health and well-being interventions across domains and populations.

Research Assistant, *University of Rochester, Human-Computer Interaction Lab* **May 2020 — May 2021**
- Employed machine learning techniques to enable remote diagnosis of Parkinson's disease via an online platform to lower barriers to healthcare.

Research Assistant*, University of Rochester, Inter-Play Lab* **June 2020 — April 2021**
- Developed a projection-based augmented reality prototype to measure its impact on seamless ASL acquisition and communication between hearing parents and deaf/hard-of-hearing children through near-object ASL video projection during play sessions.

Machine Learning Intern, *Tensorflow, Google Summer of Code*. **May 2019 — Sept 2019**
- Implemented efficient traditional machine learning algorithms in Swift for the beta Swift for Tensorflow (S4TF) framework.

Teaching and Mentoring

Teaching Assistant, *Johns Hopkins University* **Fall 2023**
- Graded and held office hours EN.601. 490/690 **Introduction to Human-Computer Interaction**
- Guest lecture on Participatory Design

Teaching Assistant, *Johns Hopkins University* **Spring 2023/2025**
- Graded and held office hours EN.601.491/691 **Human-Robot Interaction**

Student Mentor, *Johns Hopkins University* **2021-present**
- Mentored 10 undergraduate and 6 graduate student at Johns Hopkins University
- Mentees have gone on to pursue Ph.D. degrees at Carnegie Mellon Robotics Institute, Rice University Mechanical Engineering and Yale University Computer Science

Service

Lead Program Chair for ACM/IEEE HRI Pioneers Workshop **2025**

- Led the peer review and cohort selection process for the 2026 ACM/IEEE HRI Pioneers Workshop

Peer Reviewer **2022 — Present**
- Peer reviewed for conferences (HRI, CHI, ICRA) and journals (THRI, IJSR, PLOS Digital Health)

Member of Organizing Committee for AAAI Fall Symposium of AI for Aging in Place **2024**
- Served as the publicity chair for a AAAI symposium aimed at bringing together researchers working on AI technologies to support aging in place

Facilitator PhD Mentor Hour **Fall 2022-23**
- Facilitated two mentor hour sessions to connect junior and senior PhD students for peer guidance

Invited Talks

University of Chicago – Introduction to Human-Robot Interaction **Fall 2025**